

2008 JC2 Prelim II Paper 2 (Answers)

Section A

Question 1

(a)

- A exocytic / secretory / transfer / transport vesicle; R: lysosome
- B Golgi body / apparatus / complex;
- C Golgi vesicles (*plural*);
- D rough endoplasmic reticulum;

(b)

later in B than D due to processing or sequence of events;
labeled amino acids are synthesized into protein / polypeptide at the ribosomes on RER;
proteins synthesized pass through RER membrane / enter RER through membrane;
proteins / polypeptides then move / diffuse / transported within cisternae of RER;
vesicles pinched off from RER move to Golgi body;
movement of vesicles takes time / movement of vesicles by cytoskeleton is slow;
vesicles then fuse with Golgi body;

(c)

via exocytosis;
vesicle moves to cell surface membrane;
vesicle membrane fuses with cell surface membrane;
using energy from ATP / ref role of cytoskeleton / microtubules;

(d)

ATP synthesis occurs in mitochondria (org. E);
Energy from ATP is released in aerobic respiration;
Large amount of energy required;
For activation of amino acids / condensation of amino acids / formation of peptide bonds / active uptake of amino acids; (R: translation)
For movement of secretory vesicles / secretion via exocytosis / cytoskeletal movement;

Question 2

(a)(i)

centromere;

(a)(ii)

kinetochore formation on centromere;

attachment of spindle fibres on kinetochore found on centromere during metaphase;

allows for equal separation of genetic material to daughter cells during anaphase;

(b)

Fig. 2.1 – separation of homologous chromosomes to opposite poles of cell;

Fig. 2.2 – separation of sister chromatids to opposite poles of cell;

(c)

sister chromatids separated and moving towards opposite poles of cell;

sister chromatids reach the opposite poles of cell;

nuclear membranes starts to form;

invagination of plasma membrane;

cleavage furrow formation;

separating into 2 cells with 2 chromosomes each;

(d)

methotrexate – reversible competitive inhibitor;

competes with substrates for active site of enzymes in metabolic pathway;

(methotrexate presence – will bind to the active site of the enzyme and prevent purine formation)

resulting in less purines formed; (since reversible reaction → formation of purine will slow down,

Ⓜ if purine formation stops)

as purines (nucleotides) needed in DNA replication;

less replication takes place in 'S' phase interphase before start of mitotic cycle;

slowing down rate of mitosis and effect of tumor forming is suppressed;

Question 3

(a)

deletion of three consecutive bases that make up a codon/deletion of codon;

once a codon is removed from the sequence;

a.a. encoded by that codon will be left out of polypeptide chain;

and no frame shift/rest of codons unaffected;

(b)

new codon is translated into the same amino acid due to redundancy of the genetic code/wobbling;

new amino acids may have properties similar to the original amino acid;

substitution occurs in a region where the exact sequence does not affect the configuration of the active site;

(c)

each tRNA carries a specific amino acid;

enters A-site of ribosome;

with anticodon on tRNA complementary to codon on mRNA binds;

peptidyl transferase/ribozyme catalyses formation of peptide bond between next a.a. (in A site) and growing polypeptide chain (in P site);

ribosome translocates 1 – codon on mRNA and tRNA in E-site ejected;

steps are repeated till polypeptide chain completed;

(d)

loss of amino acid alters 3D configuration/tertiary structure of protein;

change in shape makes it unable to bind to the sugar molecules to form functional glycoprotein;

Question 4

(a)

a cluster of genes whose expression is controlled by a single operator;

operon consists of promoter region which RNA polymerase binds to;

operator which repressor protein binds to and structural genes that codes for the proteins;

(b)

enzymes that control this pathway in *E.coli* are coded for by genes in the *lac* operon;

which consists of three structural genes which codes for permease (carries lactose into the cell), galactosidase (converts lactose into glucose and galactose) and transacetylase;

regulator genes codes for a repressor that binds to the operator, blocking the attachment of RNA polymerase;

in the presence of lactose, allolactose binds to the repressor protein and alters its conformation / configuration, hence, inability of repressor binding to the operator;

RNA polymerase is not blocked and thus transcription of *lac* mRNA by RNA polymerase occurs;

transcription of the *lac* operon proceeds very slowly unless a second event also takes place;

attachment of a cAMP receptor protein (CRP) attach itself to the CRP binding site on the DNA which enhances the binding of RNA polymerase to the promoter;

CRP is activated by cyclic AMP, which in turn, causing rapid transcription of the *lac* operon;

(c)

C;

When glucose is scarce/ absence, there is an increase in concentration of cAMP;

cAMP binds to the allosteric site of CAP;

CAP assumes its active shape/ 3D conformation;

bind to CAP binding site;

attachment of CAP bends the DNA;

recruits RNA polymerase to bind to the promoter;

transcribe mRNA of the *lac* operon/ Z gene/ transcription at high level;

Question 5

(a)(i)

<i>genotype</i>	<i>phenotype</i>
VvCcBb	Red eyes
vvCCBB	Vermillon eyes

(a)(ii)

epistasis

(a)(iii) vv is epistatic over all the other alleles

cc is epistatic over C and B/b alleles

bb is epistatic over BB.

The alleles may be coding for enzymes that convert the amino acid tryptophan to the various pigments.

(b)(i) 1: 1: 2 ratio

(b)(ii) will deviate from the expected ratio due to crossing over.

Because C and B on the same loci → should be able to see more parental phenotypes as well.

Question 6

(a)

C and D

(b)

to regenerate NAD;

for glycolysis to proceed, so there is production of ATP by glycolysis;

under anaerobic conditions;

(c)(i)

nuclei & ribosomes;

(c)(ii)

glucose cannot diffuse across almost impermeable inner membrane of mitochondria;

glucose cannot gain entry into matrix of mitochondria;

glycolytic enzymes are not found in mitochondria;

so glucose cannot be broken down in mitochondria;

pyruvate is the end product of glycolysis;

in cytoplasm;

only pyruvate can enter mitochondria;

CO₂ produced in Krebs cycle;

and link reaction;

by decarboxylase/decarboxylation;

(c)(iii)

cyanide inhibits cytochrome oxidase;

inhibits oxidative phosphorylation/ stops ETC;

no dehydrogenation of NADH₂;

Krebs cycle stops / no decarboxylation;

but lactate is produced in cytoplasm;

not in mitochondria;

enzymes involved not inhibited by cyanide;

the production of lactate does not involve ETC/cytochrome oxidase;

(d)

inner membrane is highly folded forming cristae;

allowing more surface area for electron carriers in ETC and ATP synthase used in oxidative phosphorylation of aerobic respiration;

inner membrane is impermeable to H⁺/protons;

allowing H⁺ reservoir in intermembranal space creating H⁺ gradient between intermembranal space and matrix;

thus, containing proton pump proteins to allow H⁺ to cross membrane barrier;

Question 7

(a) CiCt cross with CiCt (insularia)

produces CiCi, CiCt and CtCt which are insularia and typica.

(b) Dorset – no pollution → lichen thrive on tree well → trees have a lighter surface
Typica is lighter in colour thus survives well on the trees with lichen on them as they are able to camouflage. Predators (birds) can't spot them easily.
Carbonaria will stand out thus will be spotted by the predators easily.

Birmingham → pollution → affects the lichen → tree surface looks darker
Carbonaria able to thrive better as they are darker and can camouflage better compared to the lighter typica.

Those that survive and reproduce are able to pass on the favourable traits and thus accounting for the frequencies stated.

Selective Pressure is the Predation

Selective advantage is the colour of the moths.

(c)(i) They should be able to mate and produce viable offsprings

(c)(ii) mutation should lead to failure in interbreeding

Section B

Question 8

(a)

Control Elements

1. divided into 2 types: trans & cis-acting elements;
2. cis-acting - non-coding regulatory DNA sequences that are bound by transcription factors to help regulate transcription of a gene;
3. core promoter, proximal control elements, distal control elements (silencers and enhancers);
4. trans-acting – are transcription factors that bind to cis-acting elements that regulate transcription of a gene;
5. specific transcription factors (activators & repressors) & basal transcription factors;

Core Promoters

6. prokaryote: -35 and -10 (Pribnow box), eukaryote: -25 (TATA box);
7. serves as recognition sites for both RNA polymerase & transcription factors;
8. interaction of transcription factors & RNA polymerase with promoter only produce basal level of transcription;

Proximal Control Elements

9. lies within 100-200 base pairs upstream from start site of transcription & promoter;
10. serve as recognition sites for transcription factors to bind & recruit RNA polymerase into general vicinity of promoter region;
11. e.g. in eukaryote: CAAT box & GC box;

Enhancers

12. located upstream or downstream of transcription start site;
13. bound by transcription factors known as activators;
14. a looping mechanism brings the bound activator to interact with the RNA polymerase or transcription factors at the TATA site → up-regulates transcription

Silencers

15. located upstream or downstream of transcription start site;

16. bound by transcription factors known as repressors;
17. results in inhibition of gene expression;
18. overlapping of binding sites for repressors & activators such that binding of repressor prevents binding of activator to enhancer;
19. binding of repressor to a transcription factor needed for recruitment of RNA polymerase & thus inhibiting binding of activator to transcription factor;

(b)

Features		(a) Prokaryotes	(b) Eukaryotes
1	Location	in nucleoid region of cytoplasm, not bounded by nucleus membrane	bounded by nucleus membrane
2	Size, complexity, composition	smaller, 0.6 to 10Mb; simple; single circular DNA; one chromosome; no distinct specific regions;	smallest eukaryotic genomes being less than 10 Mb in length, & largest over 100 000 Mb; complex; 2 or more linear chromosomes; Has specific distinct regions such as centromere & telomere;
3	Presence extra-chromosomal DNA	presence of plasmids, containing a small amount of DNA with few genes;	DNA with prokaryotic origins in organelles in mitochondria & chloroplast;
4	Coding sequence of a gene; amount of non-coding DNA	continuous; small amount e.g. promoter, operator	interrupted by introns; large amounts, most of DNA does not code for protein & RNA. e.g. introns, satellite DNA, control elements;
5	Organization of genes	Operon system; polycistronic genes; cluster of structural genes; regulated by promoter, operator, etc	monocistronic genes; each gene has its own promoter, terminator and other control sequences such as enhancers, silencers etc;
6	Levels of DNA packing	2 levels: negative and loop domain formation;	4 levels: nucleosomes, chromatin fibre (solenoid), looped domains, metaphase chromosome;

Features		(a) Prokaryotes	(b) Eukaryotes
7	Association with histones	negatively charged DNA bound to associated DNA binding proteins (positively charged);	negatively charged DNA bound to 8 histone proteins (positively charged) forming nucleosomes;
8	Origins of replication	Single ori	Multiple ori

(c)

1. mutations in cancer critical genes;
2. leading to uncontrolled/excessive cell division;
3. cancer is a multi-step process and requires multiple / accumulation of mutations;

Proto-oncogenes

4. gain-of-function mutations; only one copy of gene is mutated activating oncogene;
5. mutation to oncogene results in uncontrollable cell proliferation;
6. mutations in promoters or control elements lead to up-regulation of proto-oncogene expression;
7. mutations within proto-oncogene coding region;
8. result in a constitutively active gene product / gene product that binds to its substrate with increased affinity;

Tumor suppressor genes

9. loss-of function mutations;
10. both copies of gene need to be mutated;
11. loss of tumor suppressor genes allows loss of control and inappropriate cell cycle progression;
12. mutations in promoters or control elements; lead to down-regulation of tumor suppressor gene expression;
13. mutations within tumor suppressor gene coding region;
14. result in a constitutively inactive gene product / gene product that binds to its substrate with decreased affinity / or a non-functional gene product due to loss of protein structure;

Question 9

(a)

1. Na^+/K^+ pump on axon membrane;
2. to maintain membrane potential (potential difference across membranes);
3. via active transport requiring ATP;
4. for every 3 Na^+ out, 2 K^+ in;
5. non-voltage gated ion channels;
6. Na^+ & K^+ channels on axon membrane;
7. based on facilitated diffusion controlling movement of Na^+ & K^+ in and out of axoplasm;
8. down electrochemical gradient;
9. presence of anionic proteins in axoplasm maintains negative potential within axoplasm;
10. voltage gated ion channels (Na^+ , K^+);
11. changes in membrane potential when wave of depolarization reaches point on axon;
12. opening of Na^+ channels, entry of Na^+ into axoplasm causing depolarization of membrane;
13. influx of Na^+ into axoplasm;
14. A.P. generated when threshold potential exceeded;
15. A.P. transmitted/propagated to adjacent region of axon;
16. efflux of K^+ out of axoplasm as Na^+ channels close & K^+ channels open;
17. repolarisation occurs;

(b)

1. nerve impulse arrives at synaptic knob, causes Ca^{2+} gated-channels in presynaptic membrane to open;
2. Ca^{2+} diffuse quickly from synaptic cleft into synaptic knob;
3. influx of Ca^{2+} causes synaptic vesicle to fuse with presynaptic membrane;
4. neurotransmitter e.g. acetylcholine is released by exocytosis;
5. neurotransmitter molecules take about 0.5 ms to diffuse across synaptic cleft;

6. unidirectional movement of neurotransmitters across cleft;
7. neurotransmitter molecules bind to specific receptors on protein channels of postsynaptic membrane;
8. changes to conformation of specific receptors;
9. triggers opening of Na^+ ligand-gated channels;
10. Na^+ diffuse into postsynaptic neurone depolarising membrane;
11. giving rise to excitatory postsynaptic potential (EPSP);
12. if the EPSP exceeds threshold level, an A.P. is generated;
13. A.P. is transmitted along postsynaptic neurone;
14. mention spatial and temporal summation;

(c)

1. glucagon binds to ligand binding site on specific receptor (G protein linked receptor) on surface of liver cells / muscle cells;
2. causes conformational change to G-protein receptor;
3. G protein gets activated when GDP in α -subunit of heterotrimeric G protein is substituted with GTP;
4. activated α -subunit is separated from β & γ subunits;
5. subunits travels along plasma membrane to enzyme adenylyl cyclase;
6. activates adenylyl cyclase;
7. catalyses conversion of ATP to cAMP;
8. cAMP, binds to specific locations on protein kinase (e.g. protein kinase A);
9. activating catalytic units & enable phosphorylation of subsequent protein kinases;
10. activation of one protein kinase leads to activation of subsequent inactivated protein kinases;
11. ATP is needed in this phosphorylation cascade;
12. this occurs until glycogen phosphorylase is activated;
13. catalyse conversion of glycogen into glucose-1-phosphate (glycogenolysis);

14.release of glucose into bloodstream;

15.increases blood glucose level.